University of Asia Pacific
Department of Computer Science and Engineering
CSE 402: Operating System Lab

Lab Report: 07

Report Title: Disk Scheduling FCFS & SSTF

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1.Problem Statement:

Given the following disk request queue and initial disk head position:

- Request Queue: [3, 60, 10, 25, 30, 4]
- Initial Head Position: 41

Implement the following Disk Scheduling Algorithms:

- Shortest Seek Time First (SSTF)
- First Come First Serve (FCFS)

For each algorithm, calculate the total disk head movement and represent the request servicing sequence. Finally, compare the performance of SSTF and FCFS based on total head movement and determine which algorithm is more efficient for the given request queue.

2.Source Code:

```
queue = [3, 60, 10, 25, 30, 4]

head = 41

# SSTF

requests = queue.copy()

current = head

total_sstf = 0

sstf_sequence = [head]

while requests:

    nearest = requests[0]


    for r in requests:

        if abs(current - r) < abs(current - nearest):
```

```
            nearest = r
    movement = abs(current - nearest)
    print(current, "->", nearest, "=", movement)

    total_sstf += movement
    current = nearest
    sstf_sequence.append(current)
    requests.remove(nearest)

print("Total Movement:", total_sstf)
print("Sequence", sstf_sequence)

# FCFS
current = head
total_fcfs = 0
fcfs_sequence = [head]
print()
for r in queue:
    movement = abs(current - r)
    print(current, "->", r, "=", movement)
    total_fcfs += movement
    current = r
    fcfs_sequence.append(current)
print("Total Movement:", total_fcfs)
```

```
print("Sequence", fcfs_sequence)

# Comparison
print("\nComparison")
print("SSTF Total Movement:", total_sstf)
print("FCFS Total Movement:", total_fcfs)
if total_sstf < total_fcfs:
    print("SSTF is more efficient than FCFS.")

elif total_fcfs < total_sstf:
    print("FCFS is more efficient than SSTF.")

else:
    print("Both algorithms are equally efficient.")

print("Movement Saved:", abs(total_fcfs - total_sstf))
```

GitHub: https://github.com/Adika2003/CSE-402-Operating-System-Lab

3.Result:



4. For this disk scheduling problem, which is better: SSTF or FCFS disk scheduling? Explain with proper justification.

Answer: For the given disk scheduling problem, SSTF (Shortest Seek Time First) is better than FCFS (First Come First Serve) in terms of minimizing total disk head movement. The request queue is [3, 60, 10, 25, 30, 4], and the initial disk head position is 41.

FCFS services disk requests in the order they arrive, regardless of their distance from the current head position. For this problem, the servicing sequence is 41 → 3 → 60 → 10 → 25 → 30 → 4, resulting in a total head movement of 191 cylinders. Although FCFS is simple and fair because it follows the original request order, it can cause unnecessary disk head movement.

On the other hand, SSTF selects the request closest to the current head position at each step. Its servicing sequence is 41 → 30 → 25 → 10 → 4 → 3 → 60, resulting in a total head movement of 95 cylinders. Therefore, SSTF reduces the total head movement by 96 cylinders compared with FCFS, making it approximately 50.26% more efficient in reducing head movement for this request queue.

In conclusion, SSTF is better for this particular problem because it requires less total disk head movement than FCFS. However, FCFS guarantees servicing in arrival order, whereas SSTF may cause starvation for requests that remain far from the current head position

5.Discussion:

In this experiment, the FCFS (First Come First Serve) and SSTF (Shortest Seek Time First) disk scheduling algorithms were implemented using the given request queue [3, 60, 10, 25, 30, 4] with an initial disk head position of 41. The FCFS algorithm services requests in their original arrival order, resulting in a total disk head movement of 191 cylinders. In contrast, SSTF selects the nearest pending request at each step, resulting in a total head movement of only 95 cylinders. The results indicate that SSTF performs better than FCFS for this particular request queue because it reduces unnecessary disk head movement by 96 cylinders. However, FCFS ensures fair servicing in arrival order, whereas SSTF may cause starvation for requests located far from the current head position. Therefore, SSTF is more efficient in terms of minimizing total disk head movement in this experiment.